

Chapitre 10 : Instructions itératives

Solution des exercices

Exercice 1

```
private class BtnRpterActionListener implements ActionListener {
    public void actionPerformed(ActionEvent e) {
        int nbRep = Integer.parseInt(txtNbrep.getText());
        String reponse = "";

        for(int i = 0; i < nbRep; i++){
            reponse += txtTexte.getText() + "\n";
        }

        txtrReponse.setText(reponse);
    }
}
```

Exercice 2

```
public class Suite {

    public String genererNombrePaire2a30() {
        String suite = "";

        for(int i = 2; i <= 30; i+=2){
            suite = suite + " " + i;
        }
        return suite;
    }
}

import javax.swing.JOptionPane;

public class SuiteNombrePaire {

    public static void main(String[] args) {
        Suite maSuite = new Suite();
        JOptionPane.showMessageDialog(null, maSuite.genererNombrePaire2a30());
    }
}
```

Exercise 3

```
public class Suite {

    public String genererNombrePaire2a30() {
        String suite = "";

        for(int i = 2; i <= 30; i+=2){
            suite = suite + " " + i;
        }

        return suite;
    }

    public String genererMultiple10(){
        String suite = "";

        for(int i = 10; i <= 150; i+=10){
            suite = suite + " " + i;
        }

        return suite;
    }

}

import javax.swing.JOptionPane;

public class SuiteMultiple {

    public static void main(String[] args) {
        Suite maSuite = new Suite();

        JOptionPane.showMessageDialog(null, maSuite.genererMultiple10());
    }

}
```

Exercice 4 et 5

```
public class SuiteEntiers {
    private int entier1;
    private int entier2;

    public SuiteEntiers() {
        entier1 = 0;
        entier2 = 0;
    }

    public SuiteEntiers(int entier1, int entier2) {
        this.entier1 = entier1;
        this.entier2 = entier2;
    }

    public int getEntier1() {
        return entier1;
    }

    public void setEntier1(int entier1) {
        this.entier1 = entier1;
    }

    public int getEntier2() {
        return entier2;
    }

    public void setEntier2(int entier2) {
        this.entier2 = entier2;
    }

    public String suitePaireCroissante() {
        String suite = "";
        int debut;

        debut = entier1;
        if (debut % 2 != 0)
            debut++;

        for(int i = debut; i <= entier2; i += 2)
            suite += i + " ";
    }
}
```

```

        return suite;
    }

    public String suiteEntiereCroissante() {
        String suite = "";

        for(int i = entier1; i <= entier2; i++)
            suite += i + " ";

        return suite;
    }
}

```

Exercise 6

```

public class MultiplesEntiers {

    private int entierDepart;

    public MultiplesEntiers(int entierDepart) {
        this.entierDepart = entierDepart;
    }

    public String genererMultiples(int NombreMultiple) {
        String suite = "";

        for (int i = 1; i <= NombreMultiple; i++) {
            suite += (i * entierDepart) + " ";
        }

        return suite;
    }

    public static void main(String[] args) {
        MultiplesEntiers maSuite = new MultiplesEntiers(8);

        System.out.println(maSuite.genererMultiples(5));
    }
}

```

Exercise 7

```
import java.util.Calendar;
import java.util.GregorianCalendar;

public class Calendrier {
    private int annee;
    private int mois;

    public Calendrier() {
        this.annee = 0;
        this.mois = 0;
    }

    public int getAnnee() {
        return annee;
    }

    public void setAnnee(int annee) {
        this.annee = annee;
    }

    public int getMois() {
        return mois;
    }

    public void setMois(int mois) {
        this.mois = mois;
    }

    private int premierJour(){
        GregorianCalendar uneDate = new GregorianCalendar(annee, mois, 1);
        int premierJourDuMois = uneDate.get(Calendar.DAY_OF_WEEK);
        return premierJourDuMois;
    }

    private int nbJour(){
        GregorianCalendar uneDate = new GregorianCalendar(annee, mois, 1);
        int nombreJourDansLeMois;
        if (mois == 1)
            if (uneDate.isLeapYear(annee))
                nombreJourDansLeMois = 29;
            else
```

```

        nombreJourDansLeMois = 28;
    else
        if (mois == 0 || mois == 2 || mois == 4 || mois == 6 || mois == 7 || mois == 9 || mois == 11)
            nombreJourDansLeMois = 31;
        else
            nombreJourDansLeMois = 30;

    return nombreJourDansLeMois;
}

public String affCal(){

    String texteCalendrier = "Dim  Lun  Mar  Mer  Jeu  Ven  Sam\n";

    for(int i = 1; i < this.premierJour(); i++)
        texteCalendrier += "    ";

    for(int i = 1; i <= this.nbJour(); i++) {
        texteCalendrier += String.format("%02d ", i);
        if((i + this.premierJour() - 1) % 7 == 0)
            texteCalendrier += "\n";
        else
            texteCalendrier += "  ";
    }

    return texteCalendrier;
}
}

```

Exercise 8

```

private class BtnSeparerActionListener implements ActionListener {
    public void actionPerformed(ActionEvent e) {
        String mot = txtMot.getText();

        lblResultat.setText(String.valueOf(mot.charAt(0)));
        for(int i = 1; i < mot.length(); i++)
            lblResultat.setText(lblResultat.getText() + "-" + mot.charAt(i));
    }
}

```

Exercice 9

```
public class Palindrome {

    private String chaine;

    public Palindrome(String chaine) {
        this.chaine = chaine;
    }

    public String getChaine() {
        return chaine;
    }

    public void setChaine(String chaine) {
        this.chaine = chaine;
    }

    private String inverserChaine(String uneChaine){
        String chaineInv = "";

        for (int i = 0; i < uneChaine.length(); i++)
            chaineInv = uneChaine.charAt(i) + chaineInv;
        return chaineInv;
    }

    public boolean estUnPalindrome(){
        String chaineInv;
        chaineInv = inverserChaine(chaine);

        if(chaineInv.equalsIgnoreCase(chaine))
            return true;
        else
            return false;
    }
}
```

[illegible]


```

        case 'â' :
        case 'ä' :   chaineSansAccents += 'a';
                     break;
        case 'î' :
        case 'ï' :   chaineSansAccents += 'i';
                     break;
        case 'ô' :
        case 'ö' :   chaineSansAccents += 'o';
                     break;
        case 'ù' :
        case 'û' :
        case 'ü' :   chaineSansAccents += 'u';
                     break;
        case 'ç' :   chaineSansAccents += 'c';
                     break;
        default :    chaineSansAccents += chaineSansEspaces.charAt(i);
    }

    return chaineSansAccents;
}

public boolean estUnPalindromeG() {
    String chaineInv;
    String chaineTmp;

    chaineTmp = retirerEspacesEtAccents(chaine);

    chaineInv = inverserChaine(chaineTmp);

    if(chaineInv.equalsIgnoreCase(chaineTmp))
        return true;
    else
        return false;
}
}

```

Exercise 11

```
public class Placement {
    private int nbAnnee;
    private double taux;
    private double montant;

    public Placement(int nbAnnee, double taux, double montant) {
        this.nbAnnee = nbAnnee;
        this.taux = taux;
        this.montant = montant;
    }

    public int getNbAnnee() {
        return nbAnnee;
    }

    public void setNbAnnee(int nbAnnee) {
        if(nbAnnee > 0)
            this.nbAnnee = nbAnnee;
        else{
            this.nbAnnee = 1;
            System.out.println("Erreur");
        }
    }

    public double getTaux() {
        return taux;
    }

    public void setTaux(double taux) {
        if(taux > 0 && taux < 1.0)
            this.taux = taux;
        else{
            this.taux = 0.01;
            System.out.println("Erreur");
        }
    }

    public double getMontant() {
        return montant;
    }
}
```

```

    public void setMontant(double montant) {
        if(montant > 0)
            this.montant = montant;
        else{
            this.montant = 1;
            System.out.println("Erreur");
        }
    }

    public String obtenirValeur(){
        String affichage = "Année\tValeur du placement\n";

        for (int i = 1; i <= nbAnnee; i++)
        {
            montant = montant * (1 + taux);
            affichage += "" + i + "\t" + String.format("%.2f", montant) + " $\n";
        }

        return affichage;
    }
}

private class BtnNewButtonActionListener implements ActionListener {
    public void actionPerformed(ActionEvent arg0) {

        Placement monPlacement = new Placement(0,0,0);

        monPlacement.setMontant(Double.parseDouble(txtMontant.getText()));
        monPlacement.setTaux(Double.parseDouble(txtTaux.getText()));
        monPlacement.setNbAnnee(Integer.parseInt(txtNbAnnee.getText()));

        txtValeur.setText(monPlacement.obtenirValeur());
    }
}

```